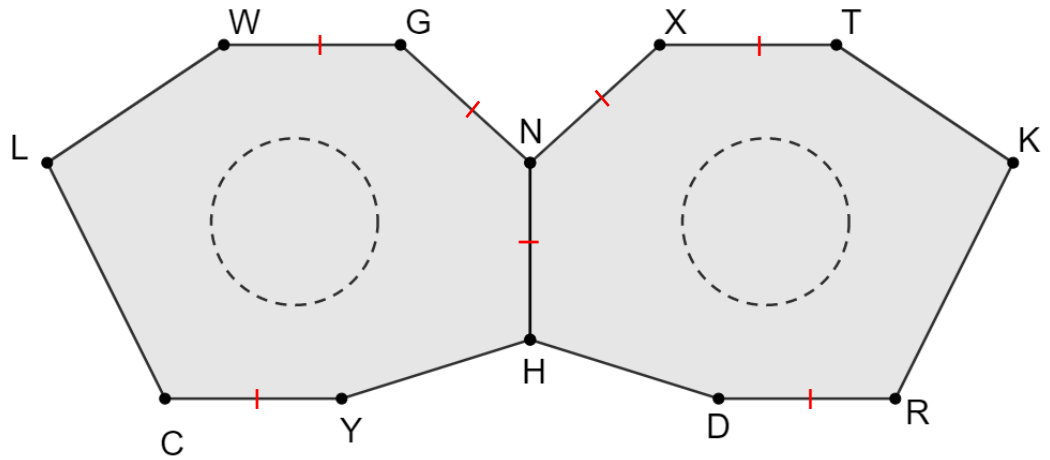


Geometry
Unit 5
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

A landscape architect is designing an area in a park that will have two fountains with benches for sitting around the fountains. The two areas are congruent polygons. The architect created a diagram where congruent polygons $LWGNHYC$ and $KTXNHDR$ represent the two areas that surround the fountains.



In polygon $KTXNHDR$, $\overline{NX} \cong \overline{NH} \cong \overline{DR} \cong \overline{XT}$. The perimeter of polygon $LWGNHYC$ is 564 inches (in.)

Some of the side measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$HD = 20x - 14.4$$

$$GW = 13x + 9.6$$

$$CL = 18x + 21.6$$

$$KT = 9x + 43.2$$

1. Write an equation that can be used to determine the value of x .

$$20x - 14.4 + 4(13x + 9.6) + 18x + 21.6 + 9x + 43.2 = 564$$

2. Determine the value of x .

$$20x - 14.4 + 52x + 38.4 + 18x + 21.6 + 9x + 43.2 = 564$$

$$99x + 88.8 = 564$$

$$99x = 475.2$$

$$x = 4.8$$

3. Determine the length of side HN .

$$HN = 13(4.8) + 9.6$$

$$= 72 \text{ inches}$$

4. Determine the length of side RK .

$$RK = 18(4.8) + 21.6$$

$$= 108 \text{ inches}$$

5. Determine the length of side XN .

$$XN \cong HN$$

$$= 72 \text{ inches}$$

Some of the angle measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$m\angle CLW = (2k + 55)^\circ$$

$$m\angle WGN = (7k - 9)^\circ$$

$$m\angle YHN = (5k + 2)^\circ$$

$$m\angle XTK = (9k - 43)^\circ$$

$$m\angle HYC = m\angle NXT + 25^\circ$$

$$m\angle KRD = m\angle TKR + 20^\circ$$

6. Given that $m\angle XTK + m\angle TKR + m\angle KRD = 360^\circ$, determine the value of k .

$$9k - 43 + 2k + 55 + 2k + 55 + 20 = 360$$

$$13k + 87 = 360$$

$$13k = 273$$

$$k = 21$$

Determine the measure of each angle.

7. $m\angle NXT = \underline{138^\circ}$
 $m\angle NXT = m\angle WGN$
 $= 7(21) - 9$
 $= 138$

8. $m\angle HYC = \underline{163^\circ}$
 $m\angle HYC = m\angle NXT + 25$
 $= 138 + 25$
 $= 163$

9. $m\angle DHN = \underline{107^\circ}$
 $m\angle DHN = m\angle YHW$
 $= 5(21) + 2$
 $= 107$

10. $m\angle LWG = \underline{146^\circ}$
 $m\angle LWG = m\angle XTK$
 $= 9(21) - 43$
 $= 146$